

12.421 / 12.621 Lecture 14

Synthetic Aperture Radar

Part 1

Topics covered

- History and overview of radar
- Basics of imaging radar
- Fundamentals of synthetic aperture radar

Overview of radar

- Originally an acronym "Radio Detection And Ranging" (Now a word)
 - Perhaps the first system we might recognize as a radar was developed by Hertz ^{in 1870s} to test Maxwell's EM theory - demonstrated that reflections from various objects were detectable
 - Military needs in the early 20th century drove significant development
 - Uses extend from detection of ships and planes to weapons development (e.g. proximity fuses)
 - Some of the first images of the ground taken from an airborne radar were produced by MIT Radiation Laboratory, precursor to Lincoln Labs
 - These early systems used a method known as side-looking aperture radar (SLAR), which "scans" the surface using a "real-aperture radar"
 - Main disadvantage to this approach is the poor azimuth resolution (as we shall see) and proportional to distance to target
 - Az. resolution inversely proportional length of antenna, "good" resolution (< 100s of meters) required huge antenna, too large to fly on aircraft or spacecraft
- In 1952
- Carl Wiley, working for Goodyear Aircraft Company, invented a method called "Doppler beam-sharpening" that synthesized a large aperture using a much smaller aperture.
 - Dramatically improved azimuth resolution $\rightarrow$ proportional to antenna length, independent of range
 - Technique now known as Synthetic aperture radar (SAR)
 - Operational SAR systems started to appear in the 1960s and continue to be extremely important to Earth science and defense and other industrial applications
 - Aside from high-resolution, SAR is valuable because it operates day and night and in virtually any weather - sees through clouds, smoke, pollution, etc

Imaging

Radar Fundamentals - How does it work?

- Radar is an active sensor - it measures range to a target by transmitting an EM signal and receiving the echo reflected by the object

- Radar instruments measure time of flight, thus range to target is simply

$$R = \frac{1}{2} V_p t \rightarrow \text{hereafter we assume air} \Rightarrow V_p \approx c$$

where the $\frac{1}{2}$ accounts for two-way travel

- Unlike optical images that are acquired by recording a lens-projected 2D image and an array of detectors, a single many detector, or chemical processes (film),

Radars form images by scanning surfaces, using the platform velocity to acquire long strip images.

* Strips often called 'swaths' or 'tracks'

- Due to strip-mapping, we distinguish directions in radar images according to the platform velocity vector:

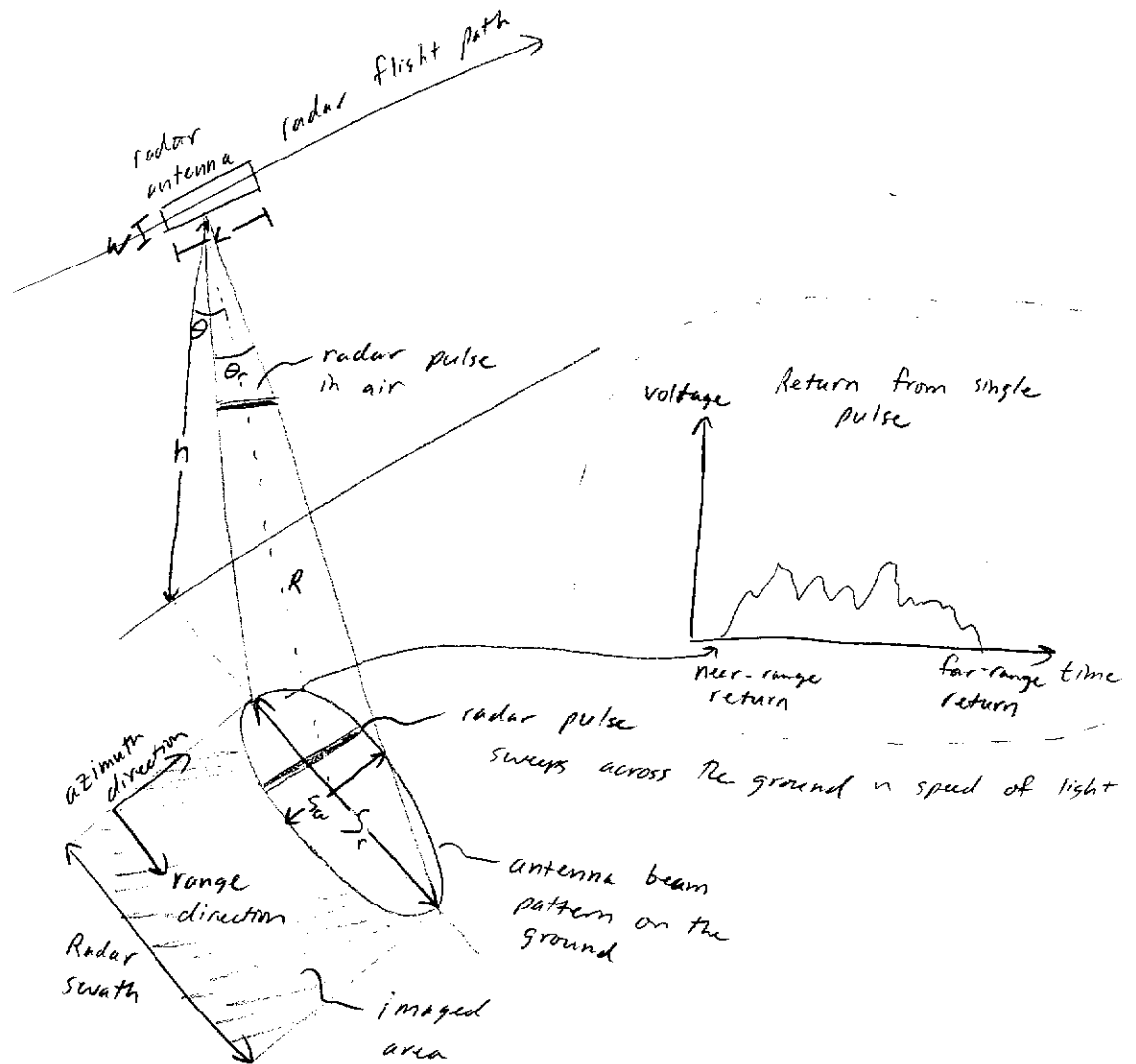
- Range (a.k.a. cross-track) is perpendicular to velocity
- Azimuth (a.k.a. along-track) is parallel to velocity

- In range direction, radar echoes are separated using the time between the echoes from different scatterers

* True for both real and synthetic apertures

- In azimuth direction, surface pixels are separated according to the angular size of the radar footprint (in real apertures) or the Doppler history (synthetic aperture)

Imaging Radar (cont.)



• Pulses transmitted from radar as platform moves

• Each pulse illuminates an ellipse on the ground

- defined in range : $\theta_r \approx \frac{\lambda}{L}$ and $S_r \approx \frac{h \theta_r}{\cos^2 \theta} = \frac{\lambda h}{L \cos^2 \theta}$
 Flat-Earth assumption

- in azimuth : $\theta_a \approx \frac{\lambda}{L}$ and $S_a \approx R \theta_a \approx \frac{\lambda h}{L \cos \theta}$

Eg. for a spacecraft: $h = 800 \text{ km}$, $\lambda = 23 \text{ cm}$ (L-band), $L = 12 \text{ m}$

$\theta = 20^\circ \rightarrow S_a = 16 \text{ km}$

Imaging Radar (cont.)

- For a real-aperture radar, $\delta_a = \lambda_a \equiv$ azimuth resolution
 - Resolution defines smallest object that can be uniquely detected
 - Note example on previous page $\rightarrow \lambda_a = 16 \text{ km}$ is horrible resolution

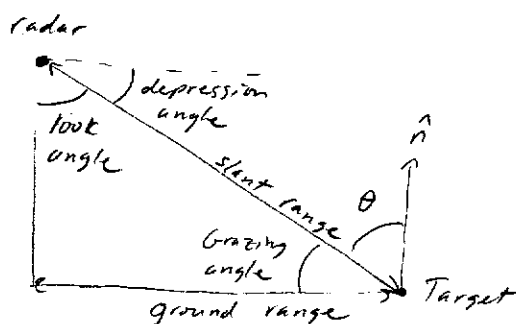
- Even if we decrease λ to 2 cm (X-band) and made $h = 200 \text{ km}$ (unreasonably low altitude for a satellite), $X = 360 \text{ m}$, which is still pretty lousy resolution for most science applications

- Poor resolution limits application of real-aperture radar for RS applications, though there are some notable exceptions $\rightarrow$ terrestrial radar

- Synthetic aperture techniques will dramatically improve ^{azimuth} resolution, and will lead to azimuth resolution equal to half the length of the antenna!!! Will discuss how in a bit

• What is range resolution?

- First, distinguish slant range and ground range



$\theta \neq$ look angle on curved planet

- slant range $\equiv$ distance from radar to scatterer (target)
- ground range $\equiv$ distance from nadir point to scatterer along a smooth (not necessarily flat) surface

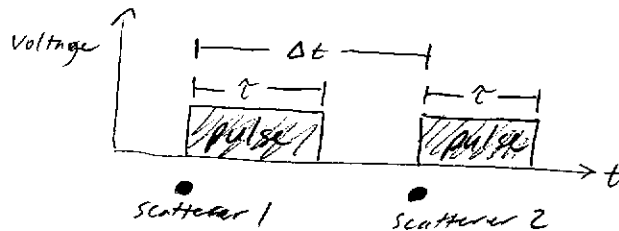
- Now consider two point targets separated by a slant range distance $x_r \Rightarrow$ echos will be separated in time

$$\Delta t = \frac{2x_r}{c}$$

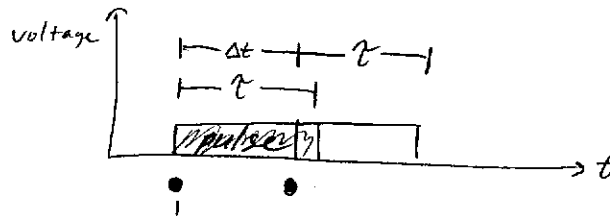
Imaging Radar (cont.)

• Range resolution (cont.)

- Radars transmit pulses, i.e. short bursts of energy
- For a pulse w/ constant frequency, two scatterers can be distinguished if the leading edge of the pulse from 2nd scatterer is received after the trailing edge of pulse from 1st scatterer



$\Delta t \geq \tau \Rightarrow$ scatterers are resolvable



$\Delta t < \tau \Rightarrow$ scatterers not resolvable for monochrome pulse

- Thus, smallest separable $\Delta t = \tau$, where $\tau \equiv$ pulse length

$$\Rightarrow \frac{2x_r}{c} = \Delta t = \tau \Rightarrow x_r = \frac{c\tau}{2}$$

- What does this mean for ground range resolution?
- Two point targets w/ ground range separation x_g

$$\Delta t = \frac{2x_g \sin \theta}{c}$$

- Based on argument above,

$$x_g = \frac{c\tau}{2 \sin \theta}$$

No dependence on distance to target!!!

- Note the dependence of ground range resolution on incidence angle $\Rightarrow$ for slopes facing radar, θ decreases $\Rightarrow x_g$ increases (and vice versa)

* ground range resolution varies dramatically in mountains

Imaging Radar (cont.)

• Range resolution (cont.)

- It would appear at this point that improving resolution is a simple matter of reducing the pulse duration τ
- Why would we not want to do that?

- A. Noise

- Recall lecture on radiation: Planck's law tells us that any material w/ $T > 0\text{K}$ emits radiation at a range of frequencies

- Radars must operate at $T > 0\text{K} \Rightarrow$ there will always be some thermal noise power density

$$P_N = \frac{k_B T \Delta f}{\lambda^2}$$

$k_B \equiv$ Boltzmann's const

$\Delta f \equiv$ bandwidth of instrument (not necessarily same as pulse bandwidth)

- Signal to noise ratio is then simply received power density P_r divided by P_N

$$\text{SNR} = \frac{P_r}{P_N}$$

- To increase SNR, want to maximize energy in each pulse $\Rightarrow$ can be done by increasing peak power or by using a longer pulse
- Peak power usually limited by power supply (esp. on satellites) $\Rightarrow$ increasing SNR depends on lengthening pulse duration
- There is a direct conflict between SNR and range resolution!
 - longer pulse = higher SNR (good), lower resolution (bad)
 - shorter pulse = lower SNR (bad), higher resolution (good)

Imaging radar (cont.)

• Range resolution (cont.)

- The solution to SNR - resolution conundrum is to abandon monochromatic pulses and to modulate frequencies instead

- 'chirp' - signal whose frequency changes in time

- Radar generally uses a linear chirp

$$\Rightarrow f(t) = f_0 + \frac{B}{\tau_p} t \quad \text{for} \quad -\frac{\tau_p}{2} \leq t \leq \frac{\tau_p}{2}$$

where

$$B = |(f + \Delta f) - f| = |\Delta f| \equiv \text{pulse bandwidth}$$

$$\tau_p \equiv \text{true pulse duration (different from effective pulse duration } \tau = 1/B)$$

- Note that $B \equiv 1/\tau$ is a property of radar instrument while τ_p is the true pulse duration

$$\ast \text{ Range resolution } \Delta x_r \propto \Delta x_g \propto \frac{1}{B}$$

$$\ast \text{ SNR scales } \propto \tau_p$$

$\ast$ Using Frequency modulated (FM) pulses, possible to construct system and pulse w/ high bandwidth ($\Rightarrow$ fine resolution) and short pulse duration τ_p ($\Rightarrow$ high SNR)

- Signal amplitude of chirped signal

$$A(t) \propto \cos\left(\omega_0 t + \frac{\pi B}{\tau_p} t^2\right) \quad \text{for} \quad -\frac{\tau_p}{2} \leq t \leq \frac{\tau_p}{2}$$

- Matched filtering allows for recovery of fine-resolution features from pulse train $\rightarrow$ range compression

Imaging radar (cont.)

• Range compression

- Echos from different scatterers will overlap in time, but
 ... FM pulse echos will have different frequencies
 $\Rightarrow$ can be separated by appropriate filtering

- Matched filters often used in radar processing

- Idea is simple: the signal received from a scatterer will be a scaled replica of the transmitted signal delayed by time $t = 2R/c$

- Therefore, simply convolve return signal $w/$ transmit signal

- To illustrate, suppose a transmit signal

$$A(t) = A_0 e^{j(\omega(t) + \phi)} \quad \text{for } -\frac{\tau_p}{2} \leq t \leq \frac{\tau_p}{2}$$

and the received signal is

$$V_r(t) = E_r A\left(t - \frac{2R}{c}\right)$$

then the compressed radar signal is the convolution

$$\begin{aligned} V_o(t) &= \int_{-\infty}^{\infty} iA^*(\xi - t) V_r(\xi) d\xi \\ &= \tau_p E_r e^{j(\omega t - 2KR)} \underbrace{\frac{\sin(\pi B(t - 2R/c))}{\pi B(t - 2R/c)}}_{\text{norm. sinc function}} \end{aligned}$$

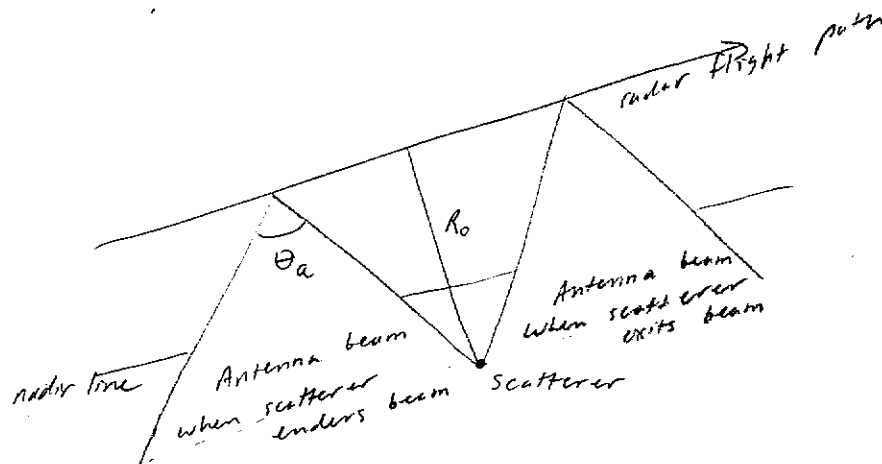
- Peak of compressed pulse occurs at $t = \frac{2R}{c}$ and it has a half power width of $\frac{1}{B}$, consistent w/ our previous observation that range resolution $\sim \frac{1}{B}$

Synthetic Aperture Radar (SAR)

imaging radar

- 'Synthetic aperture radar' utilizes data collected from a moving platform and a special processing technique that stitches together many pulses to synthesize a much larger antenna (aperture)
 - Key is to recognize that a target will be seen by the radar at different points in the radar footprint during different pulses $\Rightarrow$ slight differences in angles w.r.t. velocity vector $\Rightarrow$ Doppler shift
 - Using Doppler effect, multiple targets can be separated in the azimuth direction $\Rightarrow$ greatly improving azimuth resolution for a given radar instrument
 - Use of Doppler effect to compress signals in azimuth is the difference between SAR and real-aperture radar
 $\hookrightarrow$ range resolution and compression discussed before are the same for SAR and real-aperture radars
- * - see movie of pulse radar - *

- So, SAR relies on Doppler effect and sensor motion



Synthetic Aperture Radar (SAR) (Cont.)

- Distance between antenna and scatterer is a function of time s (a.k.a. slow time)

$$R(s) = \sqrt{R_0^2 + V^2 s^2}$$

$R_0 \equiv$ minimum distance from flight path to scatterer

$V \equiv$ platform velocity

- Usually in RS, $Vs \ll R_0$

$$\Rightarrow R_0 \approx R_0 + \frac{V^2}{2R_0} s^2$$

- The phase after range compression (see expression on previous pages) is given as

$$\phi(s) = -\frac{4\pi R(s)}{\lambda} \approx -\frac{4\pi R_0}{\lambda} - \frac{2\pi V^2}{R_0 \lambda} s^2$$

- The instantaneous frequency is simply

$$f(s) = \frac{1}{2\pi} \frac{\partial \phi(s)}{\partial s} = -\frac{2V^2}{R_0 \lambda} s \quad (s=0 \text{ when } R=R_0)$$

- Note that this $\uparrow$ is expression of linear chirp, just like we saw for FMCW pulse

- To find Doppler bandwidth $\rightarrow$ maximum time we can use for signal integration $\rightarrow$ amount of time scatterer is in antenna foot print

- As discussed before, for an antenna w/ along-track length L ,

$$\theta_a = \frac{\lambda}{L} \Rightarrow s_{\text{tot}} = \frac{\lambda R_0}{LV}$$

- Doppler bandwidth

$$B_0 = |f(s_{\text{tot}})| = \frac{2V}{L}$$

- Using match filter approach (as with range compression), the compressed signal will have a duration $1/B_0$

$\Rightarrow$ Azimuth resolution

$$\boxed{\chi_a^{\text{SAR}} = \frac{V}{B_0} = \frac{L}{2}} \rightarrow \text{Amazing! only a function of length of physical antenna!!!}$$

SAR (cont.)

- SAR Azimuth resolution

$$x_a^{SAR} = \frac{L}{2} \quad \rightarrow \text{compare w/ } x_a^{RAR} = \frac{R\lambda}{L}$$

- Smaller antenna = finer resolution? Yes because smaller antenna result in broader foot prints $\Rightarrow$ longer S_{tot} (time in beam) $\Rightarrow$ higher Doppler bandwidth
- No range dependence on azimuth resolution because longer range gives broader footprint $\Rightarrow$ higher Doppler bandwidth that cancels range effect on resolution

- Since radar is pulsed, its necessary to ensure pulse repetition frequency (PRF) sufficient to sample Doppler

$$\Rightarrow PRF \geq B_D = \frac{2V}{L}$$

$\Rightarrow$ one sample should be taken when the antenna has moved at most $\frac{L}{2}$

- Next: time, image artifacts, ambiguities and noise, then intro. to InSAR